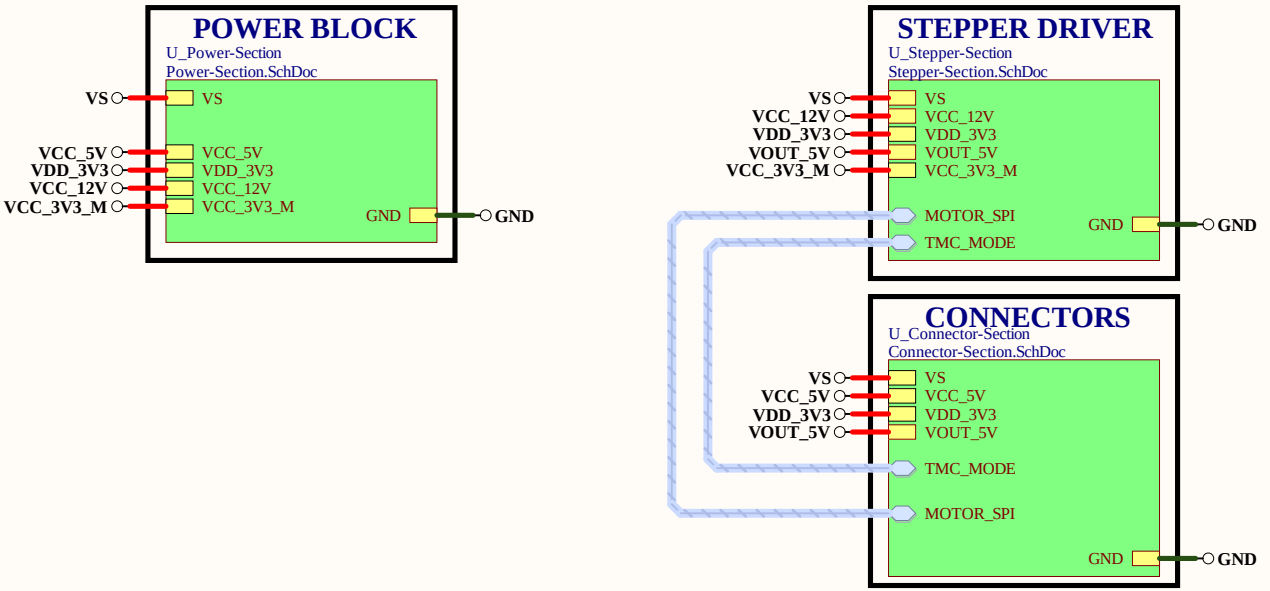
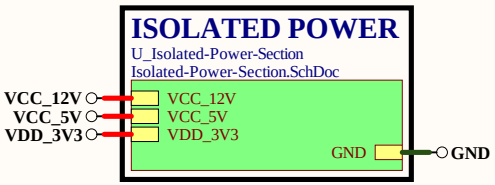
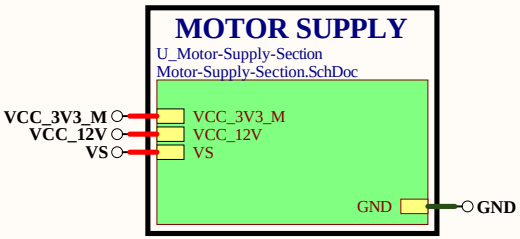
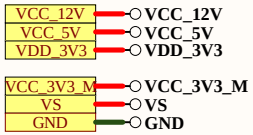
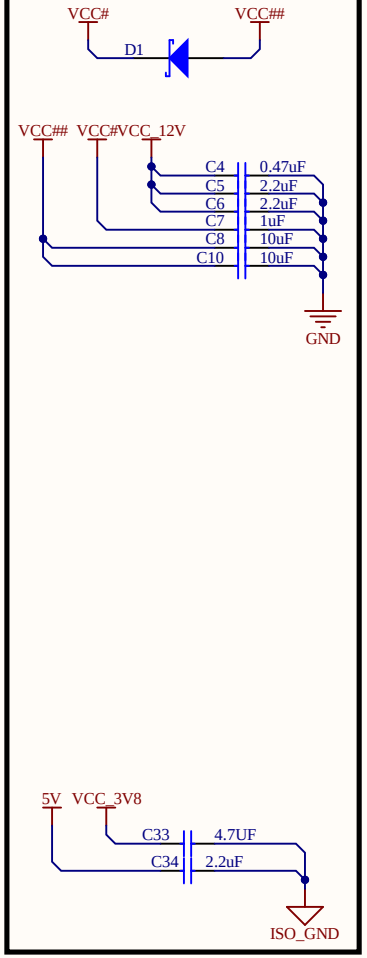

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
MOTOR_SPI
TMC_MODE
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

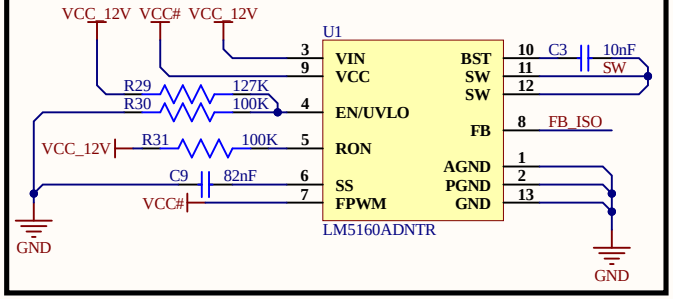

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VCC_3V3_M
VCC_12V
VS
VCC_3V3_M
VCC_12V
VS
GND
GND


VCC_12V
VCC_5V
VDD_3V3
VCC_12V
VCC_5V
VDD_3V3
VCC_3V3_M
VS
GND
VCC_3V3_M
VS
GND

DECOUPLING CAPS

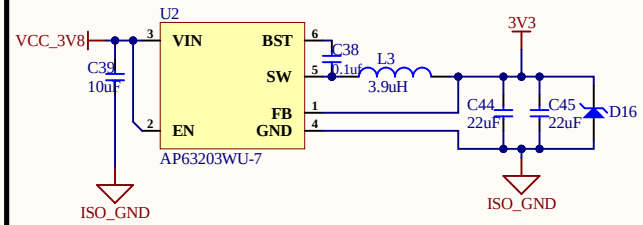


12V-3.8V ISOLATED BUCK CONVERTER

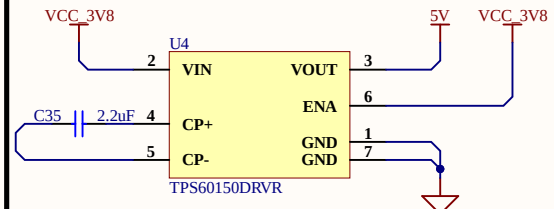


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

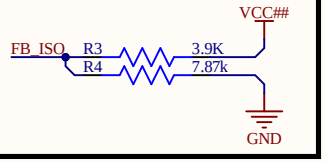
3.8-3.3V LINEAR REGULATOR



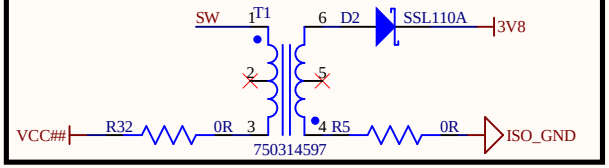
3.8-5V CHARGE PUMP



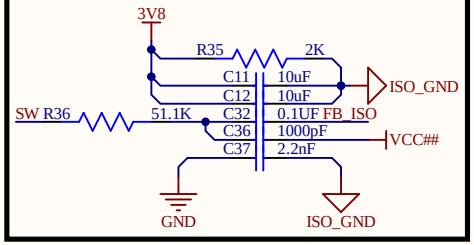
FEEDBACK NETWORK



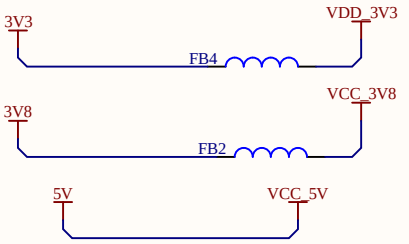
FLYBACK TRANSFORMER

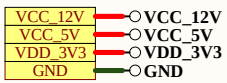


SUPPORTING CIRCUIT

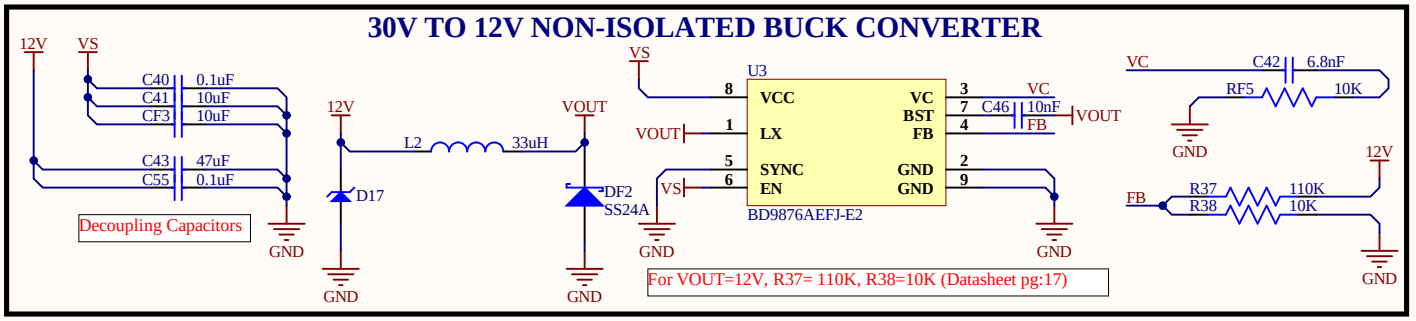


FERRITE CONNECTIONS

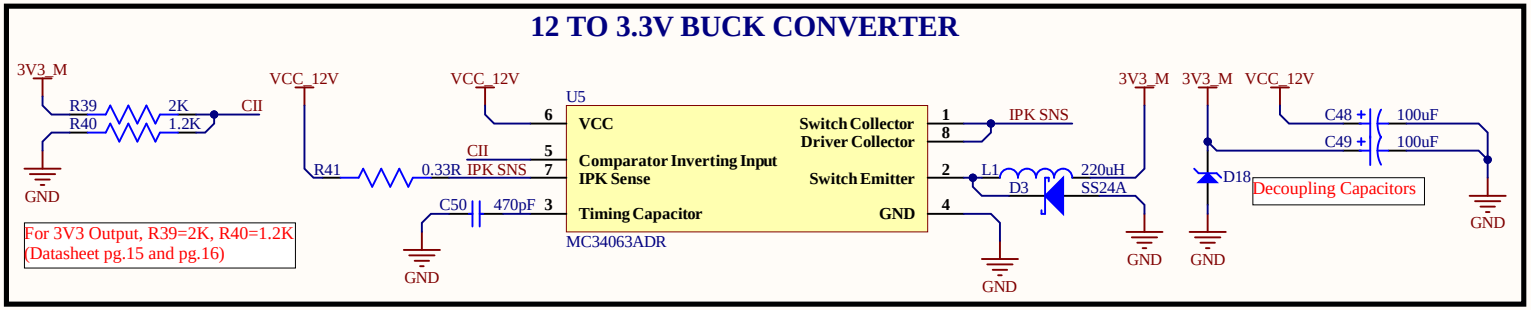




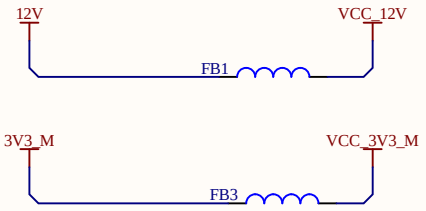
30V TO 12V NON-ISOLATED BUCK CONVERTER

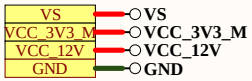


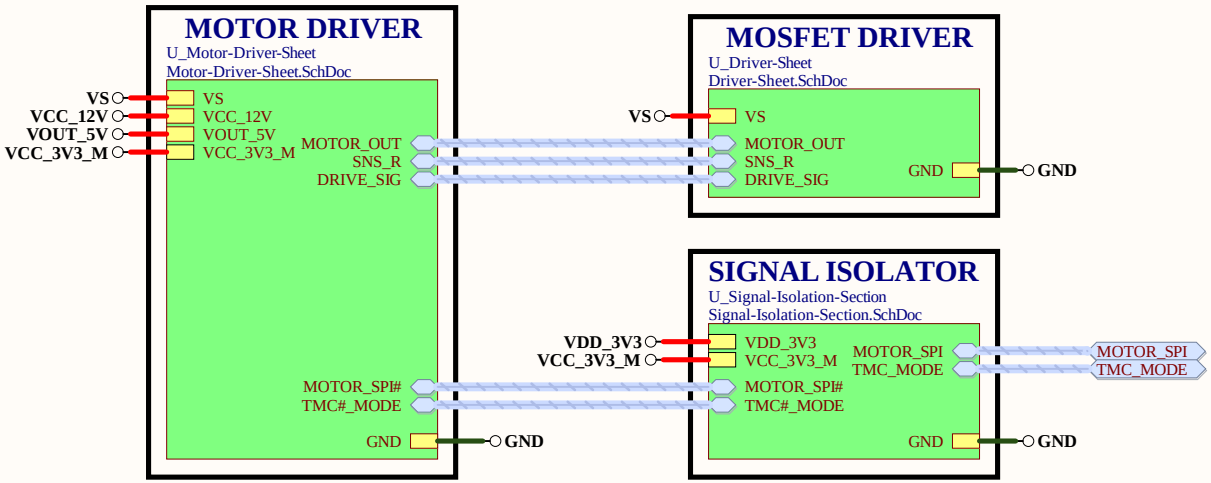
12 TO 3.3V BUCK CONVERTER

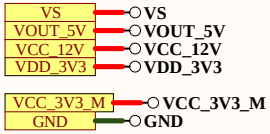


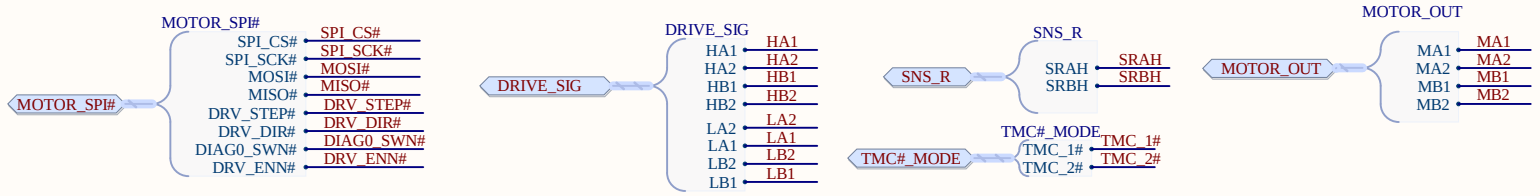
FERRITE CONNECTIONS



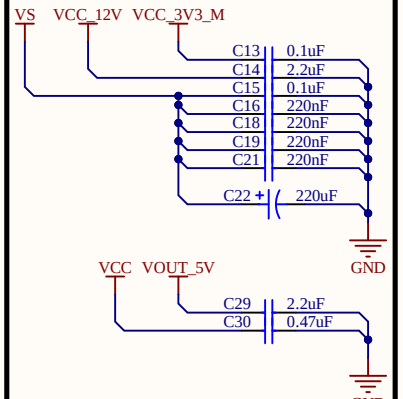



MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE


VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND

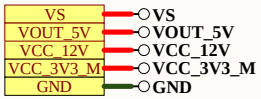


DECOUPLING CAPS

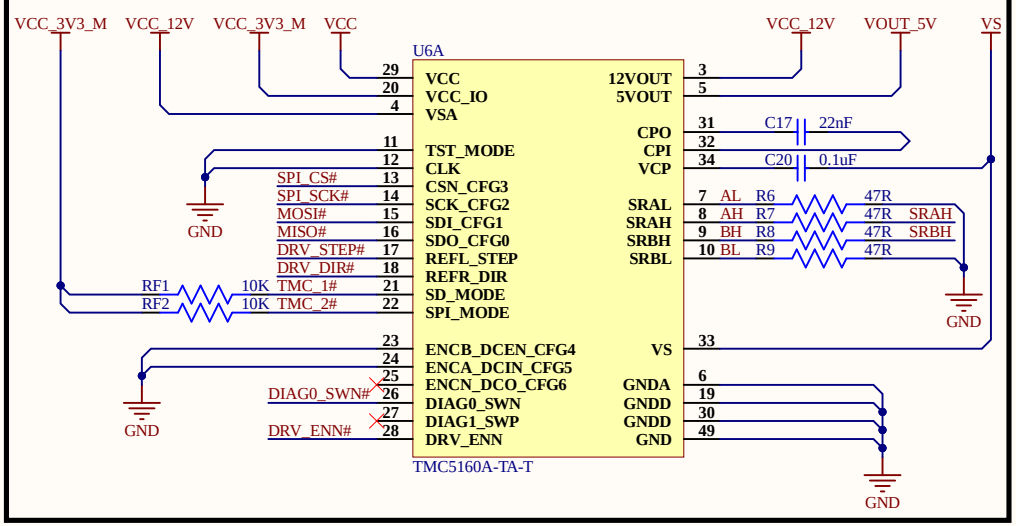


SERIES RESISTOR

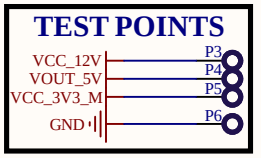




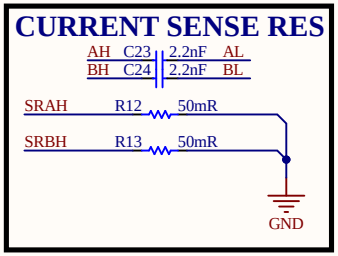
STEPPER MOTOR DRIVER-PART-A



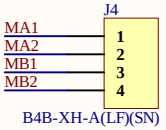
TEST POINTS



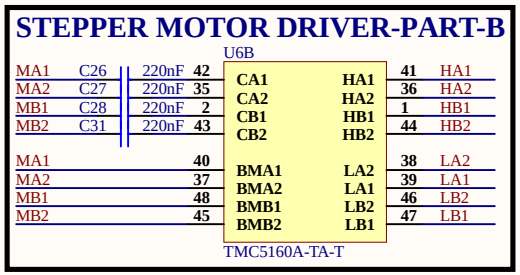
CURRENT SENSE RES



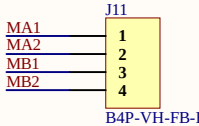
MOTOR OUTPUT

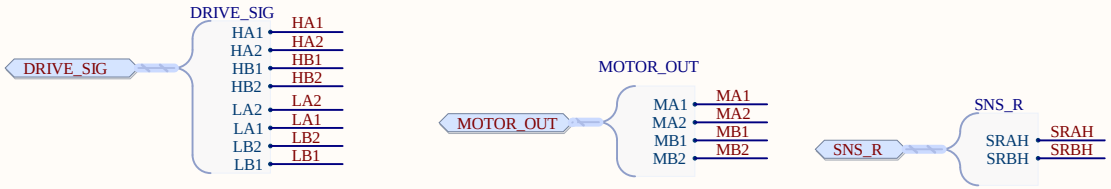


STEPPER MOTOR DRIVER-PART-B

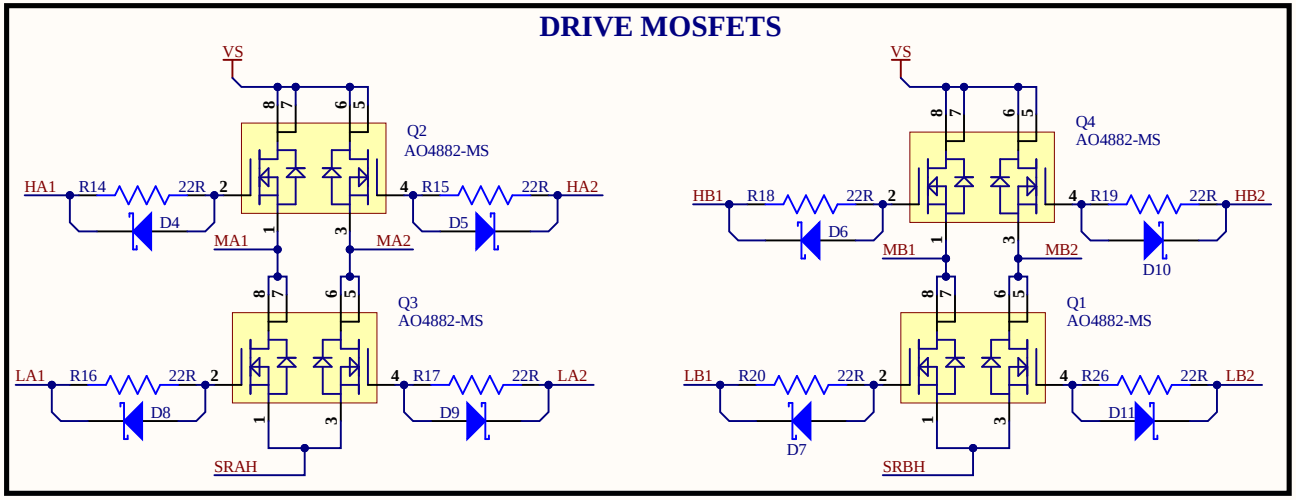


MOTOR OUTPUT FOR NEMA34

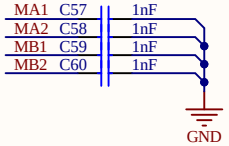




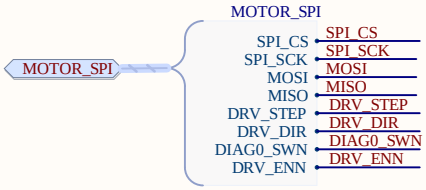
DRIVE MOSFETS

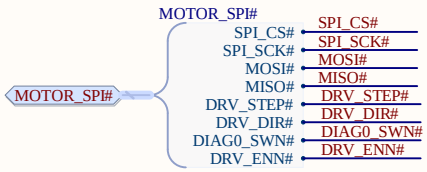


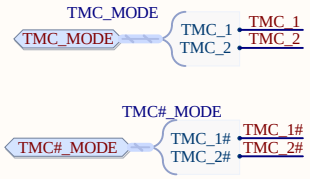
DECOUPLING CAPS

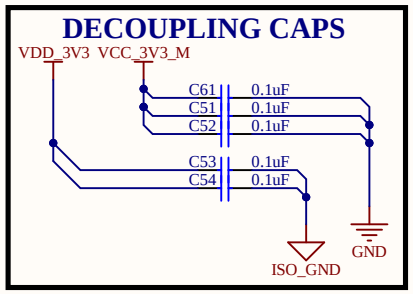


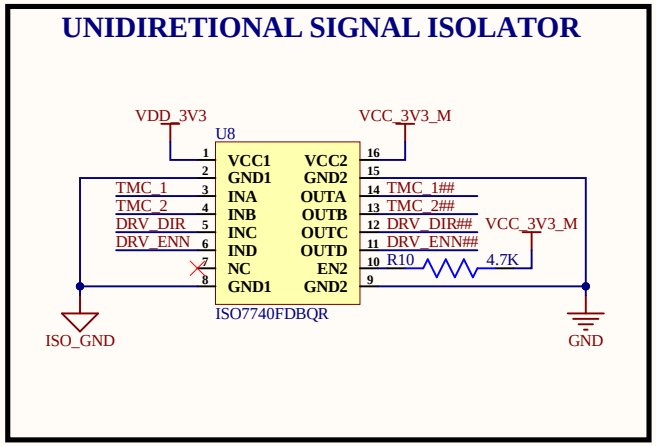


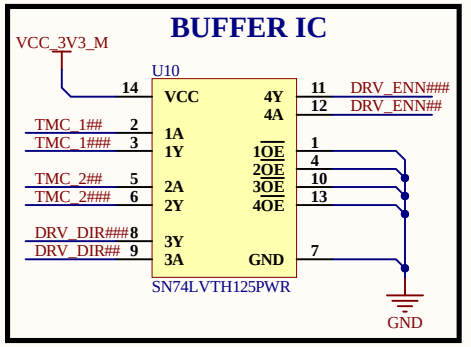


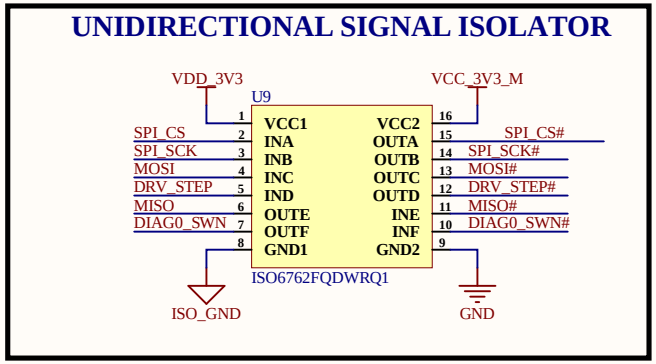


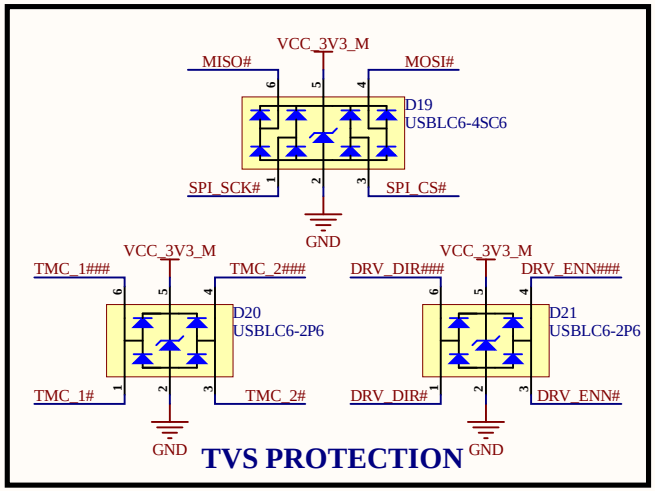






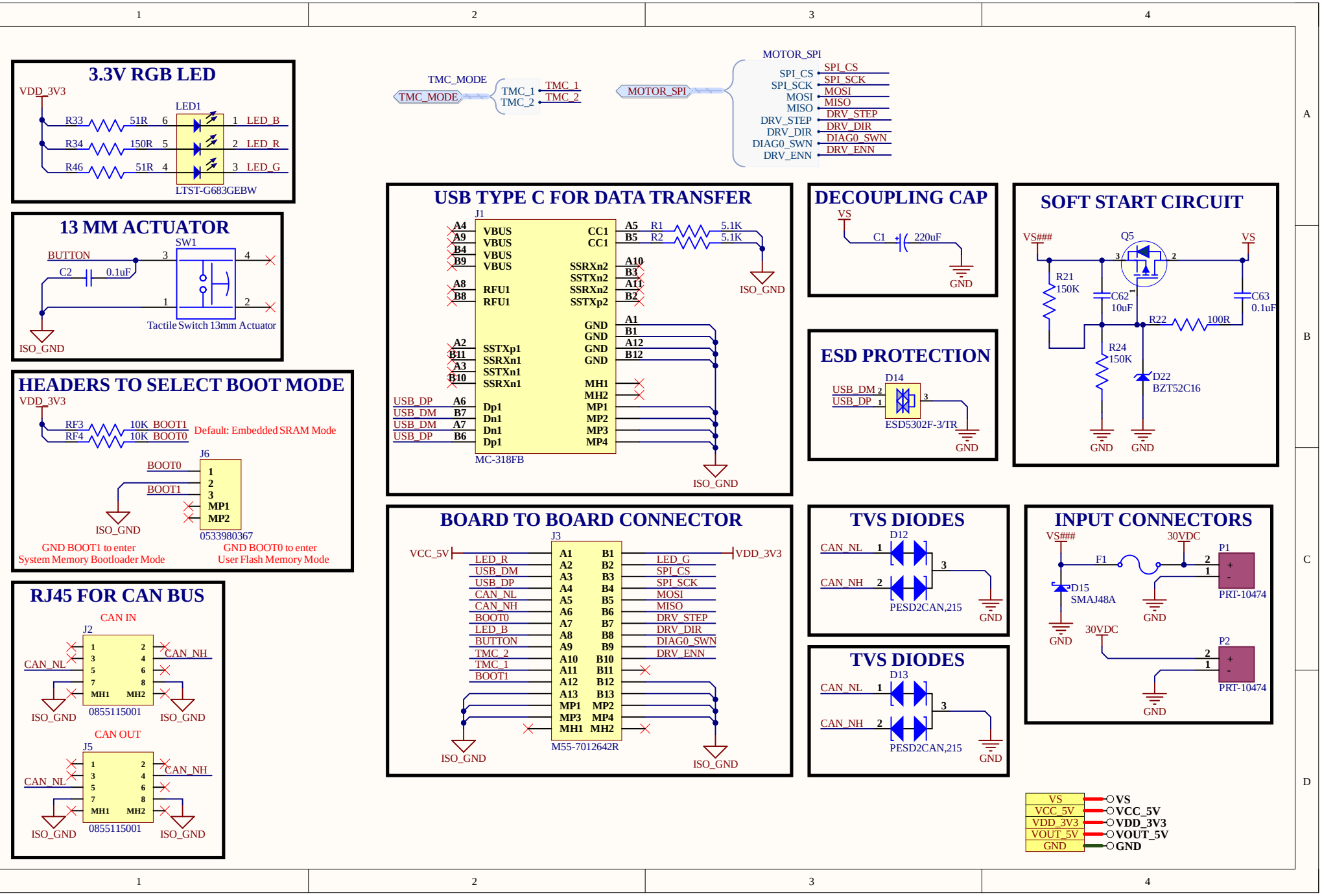








Title		
Size A4	Number	Revision
Date: 1/16/2024		Sheet of
File: C:\Users\..\Signal-Isolation-Section.SchDoc		Drawn By:


3.3V RGB LED
VDD_3V3
LED1
R33 51R 6 1 LED_B
R34 150R 5 2 LED_R
R46 51R 4 3 LED_G
LTST-G683GEBW
13 MM ACTUATOR
SW1
BUTTON
C2 0.1uF
Tactile Switch 13mm Actuator
ISO_GND
HEADERS TO SELECT BOOT MODE
VDD_3V3
RF3 10K BOOT1
RF4 10K BOOT0
Default: Embedded SRAM Mode
J6
BOOT0
BOOT1
MP1
MP2
0533980367
ISO_GND
GND BOOT1 to enter System Memory Bootloader Mode
GND BOOT0 to enter User Flash Memory Mode
RJ45 FOR CAN BUS
CAN IN
J2
CAN_NL
CAN_NH
MH1
MH2
0855115001
ISO_GND
CAN OUT
J5
TMC_MODE
TMC_1
TMC_2
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
USB TYPE C FOR DATA TRANSFER
J1
VBUS
RFU1
SSTXp1
SSRXn1
SSTXn1
SSRXn1
Dp1
Dn1
CC1
SSRXn2
SSTXn2
SSRXn2
SSTXp2
GND
MH1
MH2
MP1
MP2
MP3
MP4
R1 5.1K
R2 5.1K
USB_DP
USB_DM
MC-318FB
DECOUPLING CAP
VS
C1 220uF
GND
ESD PROTECTION
D14
ESD5302F-3/TR
SOFT START CIRCUIT
VS###
Q5
R21 150K
C62 10uF
C63 0.1uF
R22 100R
R24 150K
D22 BZT52C16
BOARD TO BOARD CONNECTOR
J3
VCC_5V
VDD_3V3
LED_R
LED_G
USB_DM
SPI_CS
USB_DP
SPI_SCK
CAN_NL
MOSI
CAN_NH
MISO
BOOT0
DRV_STEP
LED_B
DRV_DIR
BUTTON
DIAG0_SWN
TMC_2
DRV_ENN
TMC_1
BOOT1
M55-7012642R
TVS DIODES
D12
PESD2CAN,215
D13
INPUT CONNECTORS
30VDC
F1
D15 SMAJ48A
P1
P2
PRT-10474
VS
VCC_5V
VDD_3V3
VOUT_5V
GND